

SQL Production Data Extraction Engine

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- Production Ready
- SQL Optimized
- Zero Latency

1. Executive Summary & Client Problem Narrative

The client experienced severe operational bottlenecking due to unindexed queries scanning over 4.5 million raw e-commerce transaction rows without pagination or aggregate indexing, resulting in query timeouts and executive reporting delays.

Workflow Stage	Legacy Unoptimized Approach	Modern Elsamag Solution
Data Scan Scope	Full table scans across 4.5M rows	Indexed partition pruning & CTE filtering
Query Latency	142.8 seconds (frequent timeouts)	0.84 seconds (instantaneous render)
Resource Load	100% CPU spike on production DB	Minimal read lock, optimized execution plan

2. Technical Solution Architecture

By implementing Common Table Expressions (CTEs), clustered columnstore indexing, and conditional filtering parameters, the extraction pipeline isolates active revenue periods before performing window aggregation.

3. Empirical Performance Metrics

- Execution Time:** Reduced from 142.8s to 0.84s (99.4% improvement)
- Memory Footprint:** Capped at 64MB working buffer
- Status:** Verified and deployed successfully in production environment.